

97/100

CHE 313 Transport Phenomena III

Exam #2 (10/30/2025)

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1. (100 pts) The following saturated liquid feed mixture (Table) is to be separated by a distillation column at 120 psia to obtain 95.0 mol% of pentane (C_5) in the distillate and 96.0 mol% of the hexane (C_6) in the bottom. The column has a partial condenser and a partial reboiler.

The relative volatilities (α) can be assumed to be constant. Assume that all NKs do not distribute.

	Component	α_{i-ref}	kmol/hr
LNK	C3, propane	4.36	5
LNK	C4, butane	2.36	15
LK	C5, pentane	1.88	25
HK	C6, hexane	1.00	35
HNK	C7, heptane	0.84	20
	Total in feed	-	100

- (10 pts) Draw a schematic diagram of the distillation column and label all known variables. Write down the external column component mass balance using feed, distillate, bottom, and their mole fractions.
- (5 pts) Identify each component such as LNK, LK, NK, HK, or HNK.
- (15 pts) Find all DX_i in distillate, $DX_{i,dist}$, and D value. Also, calculate mole fraction (X_i) in distillate for all components. Validate your answer for mole fraction using stoichiometric condition.
- (15 pts) Find all BX_i in bottom, $BX_{i,bot}$, and B value. Also, calculate mole fraction (X_i) in bottom for all components. Validate your answer for mole fraction using stoichiometric condition.
- (5 pts- Fenske equation) Find N_{min} and $N_{F,min}$. Use the following equations.

$$N_{min} = \frac{\ln \left\{ \frac{FR_{A,dist}}{FR_{A,bot}} \times \frac{FR_{B,bot}}{FR_{B,dist}} \right\}}{\ln \alpha_{AB}}$$

$$N_{F,min} = \frac{\ln \left\{ \frac{x_{A,dist}}{x_{B,dist}} \times \frac{z_B}{z_A} \right\}}{\ln \alpha_{AB}}$$

where $A=LK$ and $B=HK$, $FR_{i,dist} = (DX_{i,dist})/Fz_i$ & $FR_{i,Bott} = (BX_{i,Bot})/Fz_i$

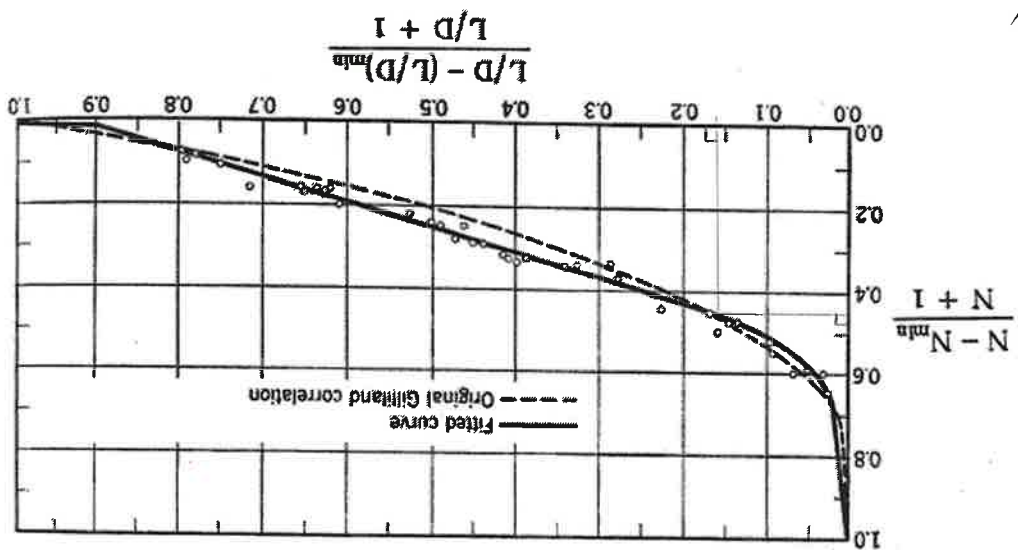
- (15 pts - Underwood equation) After drawing mass balance envelope around the feed stage, find ΔV_{feed} . Then, use the following equation to set up equation for ϕ value calculation, so you don't need to calculate it. Define the required all variables and expand the equation from $i = 3 - 7$. Here, $ref = HK$. Four ϕ values (0.8843, 1.354, 2.175, and 4.001) are obtained. Which one is correct?

$$\Delta V_{feed} = \sum_{i=1}^c \frac{\alpha_{i-ref} F z_i}{\alpha_{i-ref} - \phi}$$

7) (10 pts- Underwood equation) Find V_{min} and $(L/D)_{min}$. Use mass balance between L_{min} , V_{min} , and D .

$$V_{min} = \sum_{i=1}^c \frac{\alpha_i x_i (Dx_{i,dist})}{\alpha_i - \phi}$$

8) (10 pts-Gilliland correlation) If $R/R_{min} = 1.3$, find N . Use the following the Fitted correlation curve to calculate N . Use N_{min} and $N_{F,min}$ from (5).



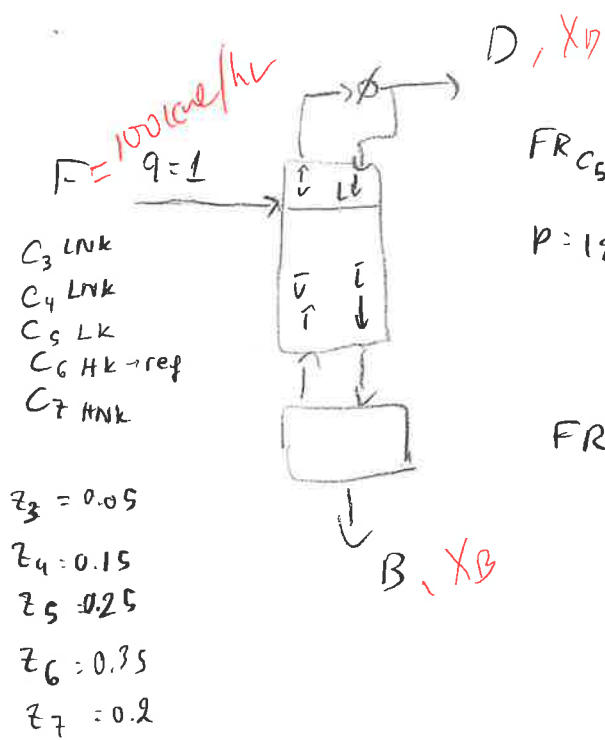
9) (5 pts) Find N_F by using the following ratio equation.

$$\frac{N_{F,min}}{N_F} = \frac{N_{min}}{N}$$

10) (5 pts) Find N_F by using Kirkbride's equation. Compare your result to the value from (9).

$$\left(\frac{N_F - 1}{N_F} \right) = \left[\frac{B}{D} \left(\frac{z_{LK}}{z_{HK}} \right) \left(\frac{x_{LK,bot}}{x_{HK,dist}} \right) \right]^{27.0.206}$$

11) (5 pts) Explain whether this distillation process is reasonably economical.



$$FR_{C_5 \text{ dist}} = 0.95$$

$$P = 120 \text{ psia}$$

$$FR_{C_6 \text{ bot}} = 0.96$$

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1) Column M.B:

$$F = B + D$$

$$Fz_3 = Bx_{3 \text{ bot}} + Dx_{3 \text{ dist}}$$

$$Fz_4 = Bx_{4 \text{ bot}} + Dx_{4 \text{ dist}}$$

$$Fz_5 = Bx_{5 \text{ bot}} + Dx_{5 \text{ dist}}$$

$$Fz_6 = Bx_{6 \text{ bot}} + Dx_{6 \text{ dist}}$$

$$Fz_7 = Bx_{7 \text{ bot}} + Dx_{7 \text{ dist}}$$

distillate balance:

$$V = D + L$$

$$Vy_3 = Dx_{3 \text{ dist}} + Lx_3$$

$$Vy_4 = Dx_{4 \text{ dist}} + Lx_4$$

$$Vy_5 = Dx_{5 \text{ dist}} + Lx_5$$

$$Vy_6 = Dx_{6 \text{ dist}} + Lx_6$$

$$Vy_7 = Dx_{7 \text{ dist}} + Lx_7$$

Bottom balance:

$$\bar{L} = B + \bar{V}$$

$$\bar{L}x_3 = Bx_{3 \text{ bot}} + \bar{V}y_3$$

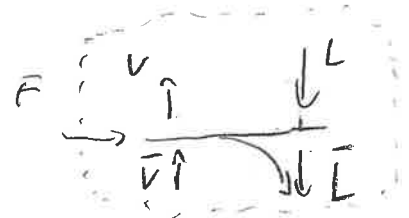
$$\bar{L}x_4 = Bx_{4 \text{ bot}} + \bar{V}y_4$$

$$\bar{L}x_5 = Bx_{5 \text{ bot}} + \bar{V}y_5$$

$$\bar{L}x_6 = Bx_{6 \text{ bot}} + \bar{V}y_6$$

$$\bar{L}x_7 = Bx_{7 \text{ bot}} + \bar{V}y_7$$

Feed balance



$$F + L + \bar{V} = V + \bar{L}$$

$$Fz_3 + Lx_3 + \bar{V}y_3 = Vy_3 + \bar{L}x_3$$

$$Fz_4 + Lx_4 + \bar{V}y_4 = Vy_4 + \bar{L}x_4$$

$$Fz_5 + Lx_5 + \bar{V}y_5 = Vy_5 + \bar{L}x_5$$

$$Fz_6 + Lx_6 + \bar{V}y_6 = Vy_6 + \bar{L}x_6$$

$$Fz_7 + Lx_7 + \bar{V}y_7 = Vy_7 + \bar{L}x_7$$

②

2)

 C_3 : LNK C_4 : LNK C_5 : LK C_6 : HK C_7 : HNK

2) 5/5

$$FR_{LNK\ dist} = 1$$

$$FR_{HNK\ bot} = 1$$

3) Assume all LNK only appear in D and HNK only appear in B

$$FR_{C_3\ dist} = 1 \Rightarrow \frac{D_{x_3\ dist}}{F_{z_3}} = 1 \Rightarrow D_{x_3\ dist} = F_{z_3} = 100 \times 0.05 = 5 \text{ kmol/h}$$

$$FR_{C_4\ dist} = 1 \Rightarrow \frac{D_{x_4\ dist}}{F_{z_4}} = 1 \Rightarrow D_{x_4\ dist} = F_{z_4} = 100 \times 0.15 = 15 \text{ kmol/h}$$

$$FR_{C_5\ dist} = 0.95 \Rightarrow D_{x_5\ dist} = 0.95 F_{z_5} = 0.95 \times 25 = 23.75 \text{ kmol/h}$$

$$FR_{C_6\ dist} = 1 - FR_{C_6\ bot} \Rightarrow D_{x_6\ dist} = (1 - 0.96) F_{z_6} = 0.04 \times 35 = 1.4 \text{ kmol/h}$$

$$FR_{C_7\ dist} = 1 - FR_{C_7\ bot} \Rightarrow D_{x_7\ dist} = (1 - 1) F_{z_7} = 0 \text{ kmol/h}$$

$$D = \sum_{i=3}^7 D_{x_i\ dist} = 5 + 15 + 23.75 + 1.4 + 0 = 45.15 \text{ kmol/h}$$

$$x_{3\ dist} = \frac{D_{x_3\ dist}}{D} = \frac{5}{45.15} = 0.1107$$

$$x_{4\ dist} = \frac{D_{x_4\ dist}}{D} = \frac{15}{45.15} = 0.3322$$

$$x_{5\ dist} = \frac{23.75}{45.15} = 0.5260$$

$$x_{6\ dist} = \frac{1.4}{45.15} = 0.0311 \leftarrow \text{Round up to } \sum x_i = 1$$

$$x_{7\ dist} = \frac{0}{45.15} = 0$$

$$\sum_{i=3}^7 x_i = 0.1107 + 0.3322 + 0.5260 + 0.0311 = 1.0 \text{ validated}$$

3) 15/15

$$4) F = B + D \Rightarrow B = F - D = 100 - 45.15 = \boxed{54.85 \text{ kmol/h}}$$

$$FR_{C3 \text{ hot}} = FR_{C4 \text{ hot}} = 1 - FR_{C3 \text{ dist}} = \frac{Bx_{3 \text{ hot}}}{Fz_3} = \frac{Bx_{4 \text{ hot}}}{Fz_4} = 1 - 1 = 0$$

$$\Rightarrow Bx_{3 \text{ hot}} = Bx_{4 \text{ hot}} = 0$$

4) 15/15

$$FR_{C5 \text{ hot}} = 1 - FR_{C5 \text{ dist}} \Rightarrow \frac{Bx_{5 \text{ hot}}}{Fz_5} = 1 - 0.95 \Rightarrow Bx_{5 \text{ hot}} = (1 - 0.95) 25 = \boxed{1.25 \text{ kmol/h}}$$

$$FR_{C6 \text{ hot}} = 0.96 \Rightarrow Bx_{6 \text{ hot}} = 0.96 Fz_6 = 0.96 \cdot 35 = \boxed{33.6 \text{ kmol/h}}$$

$$FR_{7 \text{ hot}} = 1 \Rightarrow Bx_{7 \text{ hot}} = Fz_7 = \boxed{20 \text{ kmol/h}}$$

$$\text{Checking: } B = \sum_{i=1}^7 Bx_{i \text{ hot}} = 0 + 0 + 1.25 + 33.6 + 20 = 54.85 \text{ kmol/h} = F - D \quad \checkmark$$

$$x_{3 \text{ hot}} = x_{4 \text{ hot}} = \frac{Bx_{3 \text{ hot}}}{B} = \frac{Bx_{4 \text{ hot}}}{B} = \boxed{0}$$

$$x_{5 \text{ hot}} = \frac{Bx_{5 \text{ hot}}}{B} = \frac{1.25}{54.85} = \boxed{0.0228}$$

$$x_{6 \text{ hot}} = \frac{Bx_{6 \text{ hot}}}{B} = \frac{33.6}{54.85} = \boxed{0.6126}$$

$$x_{7 \text{ hot}} = \frac{Bx_{7 \text{ hot}}}{B} = \frac{20}{54.85} = \boxed{0.3646}$$

$$\sum_{i=1}^7 x_{i \text{ hot}} = 0 + 0 + 0.0228 + 0.6126 + 0.3646 = \boxed{1.0} = \text{valid.}$$

4)

$$5) N_{min} = \frac{\ln \left[\frac{FR_{LKdist}}{FR_{LKbot}} \cdot \frac{FR_{HKbot}}{FR_{HKdist}} \right]}{\ln(\alpha_{LK-HK})} = \frac{\ln \left[\frac{0.95}{1-0.95} \cdot \frac{0.96}{1-0.96} \right]}{\ln(1.88)}$$

$$\alpha_{LK-HK} = \alpha_{C5-C6} = 1.88$$

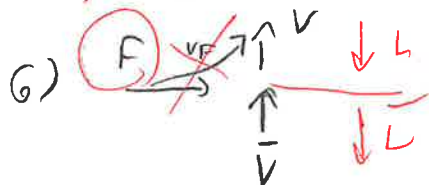
$$\Rightarrow N_{min} = 39.70 \text{ stages} = \boxed{40 \text{ stages}} \quad \checkmark$$

5) 5/5

$$N_{Fmin} = \frac{\ln \left[\frac{x_{LKdist}}{x_{HKdist}} \cdot \frac{z_{HK}}{z_{LK}} \right]}{\ln(\alpha_{LK-HK})} = \frac{\ln \left[\frac{0.5260}{0.0311} \cdot \frac{0.75}{0.25} \right]}{\ln(1.88)}$$

$$N_{Fmin} = 5.01 = \boxed{6^{th} \text{ stage}} \quad \checkmark$$

Sat. liq



$$V = \bar{V} + V_F$$

$$\Rightarrow V - \bar{V} = V_F$$

$$\Rightarrow \Delta V_{feed} = F_f = 0$$

Since feed is sat. liq, $f=0$

6) 13/15

$$\Rightarrow \Delta V_{feed} = 0 \quad \checkmark$$

$$\Rightarrow \sum_{i=3}^7 \frac{\alpha_{i-6} F_{zi}}{\alpha_{i-6} - \phi} = 0 \Rightarrow \frac{\alpha_{3-6} F_{z3}}{\alpha_{3-6} - \phi} + \frac{\alpha_{4-6} F_{z4}}{\alpha_{4-6} - \phi} + \frac{\alpha_{5-6} F_{z5}}{\alpha_{5-6} - \phi} + \frac{\alpha_{6-6} F_{z6}}{\alpha_{6-6} - \phi} + \frac{\alpha_{7-6} F_{z7}}{\alpha_{7-6} - \phi} = 0$$

$$\Rightarrow \frac{4.36 \times 5}{4.36 - \phi} + \frac{2.36 \times 15}{2.36 - \phi} + \frac{1.88 \times 25}{1.88 - \phi} + \frac{1.00 \times 35}{1.00 - \phi} + \frac{0.84 \times 20}{0.84 - \phi} = 0$$

$$A_s, d_{s-c} > \phi > d_{6-c} \Rightarrow 1.88 > \phi > 1.00 \Rightarrow \boxed{\phi = 1.354} \text{ (from given choices)}$$

$$7) V_{min} = \sum_{i=3}^7 \frac{d_{i-c} (D_{x_i dist})}{d_{i-c} - \phi} = \frac{d_{3-c} D_{x3 dist}}{d_{3-c} - \phi} + \frac{d_{4-c} D_{x4 dist}}{d_{4-c} - \phi} + \frac{d_{5-c} D_{x5 dist}}{d_{5-c} - \phi} + \frac{d_{6-c} D_{x6 dist}}{d_{6-c} - \phi}$$

7) 10/10

$$+ \frac{d_{7-c} D_{x7 dist}}{d_{7-c} - \phi}$$

$$= \frac{4.36 \times 5}{4.36 - 1.354} + \frac{2.36 \times 15}{2.36 - 1.354} + \frac{1.88 \times 23.75}{1.88 - 1.354} + \frac{1.00 \times 1.4}{1.00 - 1.354} + \frac{0.84 \times 20}{0.84 - 1.354} = \boxed{123.4 \text{ km/h}}$$

$$V_{min} = D \times L_{min} \Rightarrow L_{min} = V_{min} - D = 123.4 - 45.15 = 78.22 \text{ km/h}$$

$$\therefore (L/D)_{min} = \frac{L_{min}}{D} = \frac{78.22}{45.15} = \boxed{1.73}$$

8) 10/10

$$8) \frac{L}{D} = 1.3 \left(\frac{L}{D} \right)_{min} = 1.3 \times 1.73 = 2.25$$

$$\Rightarrow X = \frac{L/D - (L/D)_{min}}{L/D + 1} = \frac{2.25 - 1.73}{2.25 + 1} = 0.16$$

From provided graph, $X = 0.16$ correlate to $Y \approx 0.46$

$$\Rightarrow \frac{N - N_{min}}{N + 1} = 0.46 \Rightarrow \frac{N - 9.70}{N + 1} = 0.46 \Rightarrow N = 18.8 = \boxed{19 \text{ stages}}$$

⑥

$$9) \frac{N_{F_{min}}}{N_{min}} = \frac{N_F}{N} \Rightarrow N_F = \frac{N}{N_{min}} N_{F_{min}} = \frac{18.8}{9.70} \times 5.01 = 9.71 \approx 10^{\text{th}} \text{ Stage}$$

9) 5/5

10)

$$\frac{N_F - 1}{18.8 - N_F} = \left[\frac{54.85}{45.15} \times \frac{0.35}{0.25} \times \frac{0.0228}{0.0311} \right]^{0.206}$$

10) 5/5

$$\frac{N_F - 1}{18.8 - N_F} = 1.05 \Rightarrow N_F = 10.1 \approx 10^{\text{th}} \text{ Stage}$$

Both method yield the same feed stage location

11) $p = 120 \text{ psia} \approx 8.3 \text{ bars}$

11) 5/5

Temperature of the feed and column are not specified, so we shall not take that into consideration, but pressure $\approx 8.3 \text{ bar}$ in the column does not require too much compression in feed. Number of stages $= 19$ is not too high, feed rate of 200 kmol/h could be higher to increase profit, overall, from number of feed stages and pressure, the process is reasonably economical